

Mapping cell-cell interactions in the Glioblastoma Tumor Microenvironment at Spatial Resolution

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Abstract

Glioblastoma (GBM) is the most aggressive primary brain tumor, with a median survival of 15-16 months (Fekete et al., 2023) despite multimodal interventions. Using integrated single-cell and spatial transcriptomics, I mapped cell-cell interactions in the GBM tumor microenvironment (TME) at the spatial resolution. Spatial deconvolution using Cell2location revealed that the tumor cells formed a spatially restricted core domain, while CD8-T cells – despite high regional abundance – are strongly excluded from the tumor neighborhood. Neighborhood enrichment analysis confirmed the strongest negative spatial co-enrichment between tumor and CD8-T cells ($z = -51.33$) alongside pronounced CD8-T cell self-clustering ($z = 73.95$), consistent with an active immune exclusion phenotype. Ligand-receptor analysis identified JAG1 | NOTCH1 and WNT5A | STAT3 as significant tumor-to-microglia reprogramming interactions, and SFRP2 | FZD6 as a Wnt-mediated tumor-to-CD8-T cell suppression axis – with SPP1 | TFRC emerging as the most highly expressed interaction across both immune populations. Together, these findings demonstrate that the GBM TME is spatially organized into an immunosuppressive niche that physically excludes cytotoxic T cells while actively reprogramming resident microglia through Notch and Wnt pathway crosstalk.

Introduction

Glioblastoma (GBM) is an aggressive and deadly brain malignancy, with a median survival of 15-16 months despite multimodal treatment comprising surgical resection, temozolomide chemotherapy, and radiation therapy (Wen et al., 2020). The poor prognosis of GBM is driven by two interconnected features: extreme intra-tumoral heterogeneity, where malignant cells co-exist in distinct transcriptional states including NPC-like, OPC-like, AC-like, MES-like, and MES-hypoxia subtypes – defined by differences in genetic lesions, epigenetic programs, and gene expression profiles (Neftel et al., 2019; Richards et al., 2021); and a complex tumor microenvironment (TME) populated by tumor-associated macrophages (TAMs), microglia, T cells, endothelial cells, and oligodendrocytes that collectively suppress anti-tumor immunity and promote therapeutic resistance by remodeling the TME (Yoshimoto et al., 2025).

Recent advances in single-cell RNA sequencing (scRNA-seq) and spatial transcriptomic (ST) studies have showcased that the spatial organization of GBM cell states hold prognostic significance beyond cellular composition. A recent study by Zheng et al. (2023) showed that clustering of astrocyte-like (AC-like) tumor cells predict worse survival independently of their abundance, while Yoshimoto et al. (2025) identified 13 ECM ligand-receptor pairs – including SPP1-CD44 and FN1-CD44 – that are spatially restricted to specific tumor regions and associate with poor prognosis. However, these studies rely on open-source datasets with limited spatial validation, use indirect deconvolution methods, and have not made attempts to resolve the heterogeneity of tumor-associated macrophages at the subtype level.

Here I present an integrated single-cell and spatial transcriptomics analysis of patient matched GBM samples. Using scRNA-seq data from three primary and one recurrent GBM patients integrated with scVI, I constructed a cell type atlas resolving three TAM subtypes. Cell2location spatial deconvolution of a Visium CytAssist slide, followed by neighborhood enrichment, co-occurrence, and ligand-receptor analyses, revealed a spatially organized immunosuppressive niche that actively excludes cytotoxic T cells and reprograms resident microglia through Notch and Wnt pathway crosstalk.

Datasets

The raw sequencing reads in the form of FASTQ Files were retrieved from the NCBI Gene Expression Omnibus (GEO) under accession number [GSE214966](#). The dataset consists of scRNA-seq reads from 3 primary tumor tissue and from 1 recurrent tumor tissue sample from patients diagnosed with GBM. The spatial transcriptomics data was obtained from the 10x Genomics public dataset repository and is titled Visium CytAssist FFPE gene and protein expression dataset for human glioblastoma. This dataset was generated using 10x Genomics Visium CytAssist platform on a formalin-fixed paraffin-embedded (FFPE) GBM tissue section and includes both gene expression and antibody capture (protein) modalities. All data processing and alignment was performed against the GRCh38-2024-A human reference transcriptome provided by 10x Genomics.

Methodology

Data acquisition and upstream processing

To process both the scRNA-seq and ST data, a Nextflow pipeline ([GitHub](#)) was developed to automate all the major steps of the upstream analysis. The workflow integrated quality control steps using FastQC and MultiQC, alignment and UMI counting for scRNA-seq samples using Cell Ranger, spatial barcode detection and alignment for the Visium CytAssist slide using Space Ranger – all executed on the IU Quartz HPC cluster using Apptainer containers for reproducibility. Cellbender was applied to all scRNA-seq samples to remove ambient RNA contamination prior to downstream analysis.

scRNA-seq Processing, Integration, and Cell Type Annotation

Doublets were removed using Scrublet (Wolock et al., 2019) and low-quality cells were filtered based on minimum gene count (500) and mitochondrial RNA content (>20%), followed by batch integration across 4 patient samples using scVI (Gayoso et al., 2022) – a variational autoencoder that produces a batch-corrected latent representation used for Leiden clustering and UMAP visualization. Cell types were annotated manually using canonical GBM marker genes such as EGFR and SOX2 for tumor cells, TMEM119 and P2RY12 for microglia. The myeloid cluster was further sub-clustered at higher resolution to resolve three TAM subtypes – TAM_APOE, TAM_macrophage, and Monocyte – based on differential expression of APOE, SPP1, and CD14 respectively.

Spatial Transcriptomics Quality Control and Deconvolution

Spatial spots were filtered to retain only in-tissue spots with greater than 500 detected genes and less than 25% mitochondrial RNA content. Cell2location (Kleshchevnikov et al., 2022) was used for Bayesian spatial deconvolution – first training a RegressionModel on the annotated scRNA-seq data to learn cell-type specific expression signatures, then mapping these signatures onto filtered Visium spots to estimate cell type abundances. The 5th percentile of the posterior distribution was used as a conservative estimate of cell abundance per spot. Preliminary spatial deconvolution results presented prior to this report were generated from a Cell2location checkpoint trained on an unfinalized annotation and applied to unfiltered spatial spots. Final results reported here were produced by retraining Cell2location on the corrected scRNA-seq annotation and applying deconvolution exclusively to QC-filtered Visium spots, improving both the reliability of cell type signatures and the quality of spatial abundance estimates.

Spatial Neighborhood and Co-occurrence analysis

A grid-based spatial neighborhood graph with 6 neighbors per spot was constructed using Squidpy (Palla et al., 2022), and neighborhood enrichment analysis was conducted to test whether cell type pairs co-localize more than expected by chance. Spatial co-occurrence probability was computed to quantify how the likelihood of observing two cell types together changes as a function of distance, focusing on Tumor and CD8-T cell interactions to characterize the spatial exclusion phenotype. Centrality scores – degree, closeness, and average clustering – were calculated per cell type to identify which populations occupy the most central positions in the spatial interaction network.

Ligand-Receptor and Spatially Variable Gene Analysis

Ligand-receptor interactions were identified using the Squidpy CellPhoneDB permutation framework (Palla et al., 2022) with 1000 permutations, designating Tumor cells as transmitters and CD8 T cells and Microglia as receivers, with the top 20 most significant pairs selected at alpha of 0.05. Spatially variable genes were identified with Moran's I spatial autocorrelation statistic and the top hits were visualized on the tissue section to identify genes restricted to specific tumor regions. All spatial analyses were performed on the latest Cell2location-deconvolved AnnData object using dominant cell type per spot assignments based on highest abundance estimates.

Results

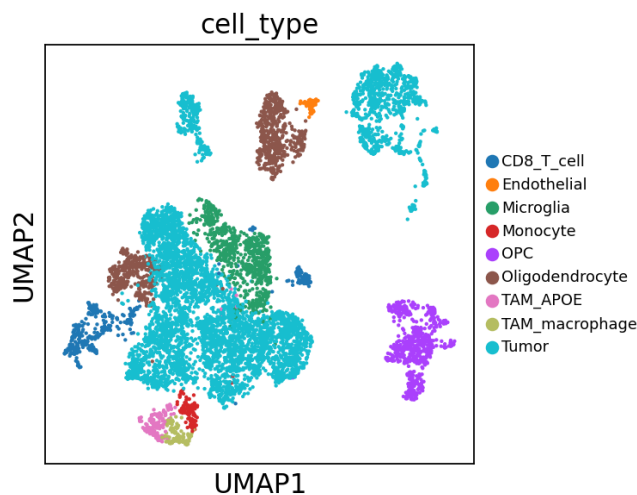


Figure 1: Single-cell transcriptomic atlas of GBM TME

scRNA-seq data from 3 primary and 1 recurrent GBM patients were integrated using scVI to produce a batch-corrected latent representation used for Leiden clustering and UMAP visualization. Integration was performed to remove tissue/donor-specific differences across the samples and to preserve cell-type specific variation. Nine distinct cell types were identified through manual annotation based on canonical marker gene expression, including tumor cells, microglia, CD8-T cells, oligodendrocytes, OPCs, TAM subtypes and endothelial cells. The myeloid cluster, known for its transcriptional heterogeneity in the GBM TME was further sub-clustered to resolve three distinct myeloid subpopulations – TAM_APOE, TAM_macrophage, and Monocyte – based on differential expression of APOE, SPP1, and CD14 respectively. Tumor cells comprised the largest population forming multiple interconnected clusters, consistent with their transcriptional heterogeneity. TAM_macrophage comprised a notably small sub-cluster relative to TAM_APOE, consistent with its low per-spot abundance observed in subsequent spatial-deconvolution. Immune populations including microglia, CD8-T cells, and TAM subtypes were clearly separated in the UMAP space, confirming successful cell type resolution across the four patients.

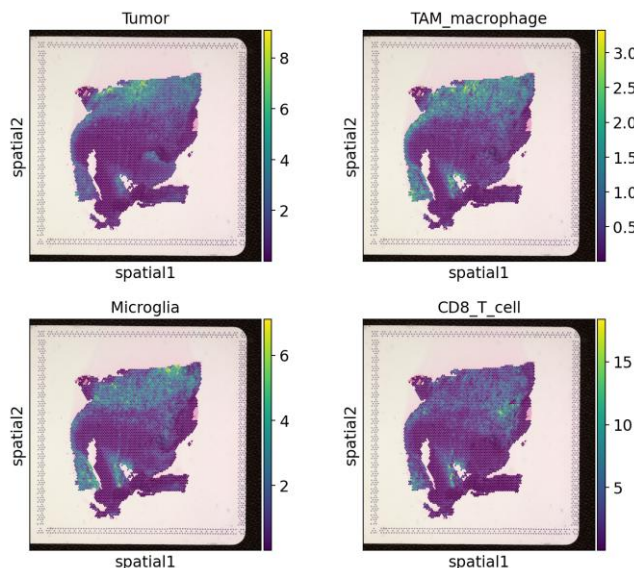


Figure 2: Cell2location reveals spatially restricted tumor core and CD8 T cell exclusion

Spatial deconvolution using Cell2location (Kleshchevnikov et al., 2022) mapped the abundance of each cell type across all Visium spots, revealing a spatially restricted tumor core predominantly occupying the upper portion of the tissue section. CD8-T cells showed unexpectedly high regional abundance (up to 15 estimated cells per spot), forming dense clusters adjacent to but spatially distinct from the tumor core. TAM_macrophage exhibited a diffuse-low abundance distribution across the tissue,

rarely achieving dominant abundance in any single spot, suggesting a dispersed infiltration pattern rather than focal co-localization with the tumor. Microglia showed a broad distribution across the entire tissue section, consistent with their role as brain-resident immune cells. Together, these patterns suggest that while myeloid populations are uniformly distributed, the CD8-T cell populations are spatially organized into discrete clusters that border but do not penetrate the tumor core.

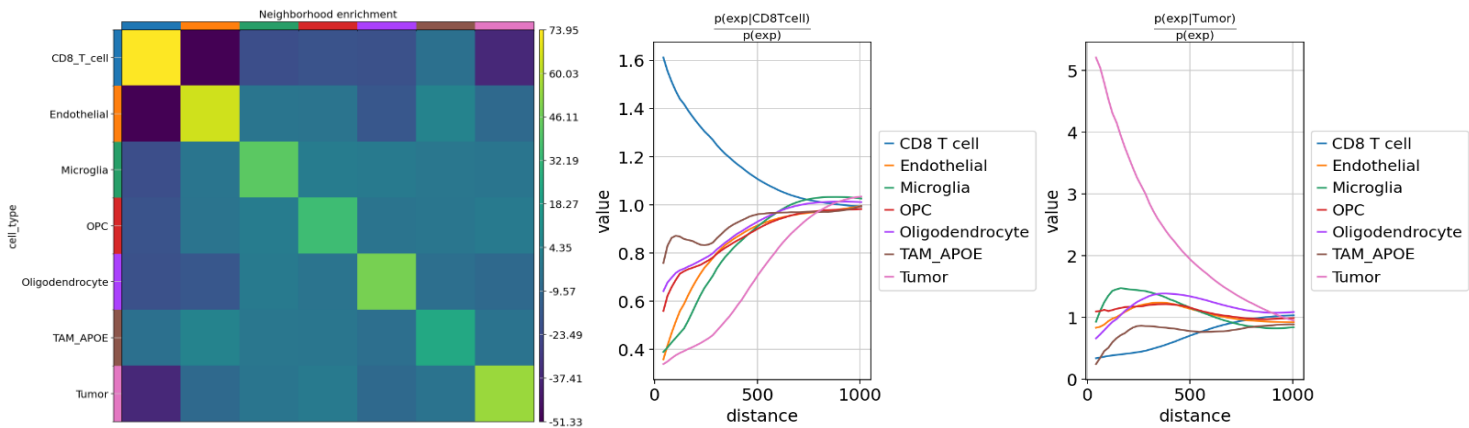


Figure 3: Neighborhood enrichment and spatial co-occurrence analysis confirms non-random organization of GBM TME

Neighborhood enrichment analysis revealed that CD8-T cell populations show the strongest positive spatial enrichment (z-score = 73.95), forming the most tightly clustered immune population in the TME, while exhibiting the strongest negative co-enrichment with tumor cells (z-score = -51.33) – indicating active spatial exclusion of cytotoxic T cells from the tumor domain. Tumor cells showed pronounced self-clustering, consistent with a cohesive tumor core that maintains spatial boundaries against immune infiltration. Normal brain populations – OPC, oligodendrocyte, and microglia – showed moderate positive self-enrichment reflecting preserved tissue architecture in the peri-tumoral region. TAM_APOE showed moderate self-enrichment, suggesting spatial clustering of the immunosuppressive TAM state. Spatial co-occurrence analysis revealed that tumor cells exhibit the strongest self-co-occurrence at short distances (~5.3), confirming a dense cohesive core, while CD8-T cells show the lowest co-occurrence probability with tumor at all distances below 600 μm – independently corroborating the spatial exclusion identified by neighborhood enrichment.

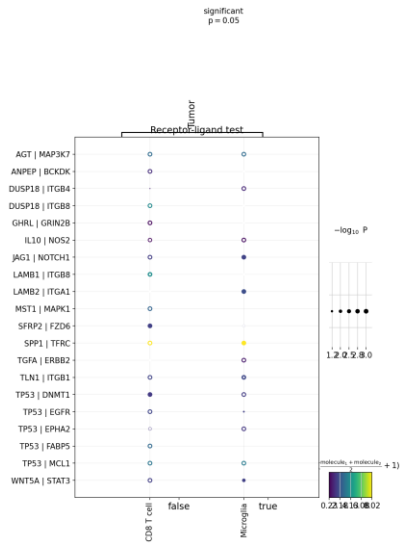


Figure 4: Ligand-receptor analysis identifies Notch, Wnt, and SPP1 signaling axes driving microglia reprogramming and CD8 T cell suppression in the GBM TME

Ligand-receptor interaction analysis (1000 permutations, $p < 0.05$) identified significant paracrine interactions between tumor cells and CD8-T cells and microglia - the two dominant immune populations in the deconvolved spatial data. SPP1 | TFRC emerged as the most highly expressed interaction across both cell types, with SPP1 (osteopontin) secreted by tumor cells signaling to the transferrin receptor on immune cells – an axis linked to iron metabolism-dependent immune suppression and

consistent with SPP1's established role in GBM immunosuppression (Lao et al., 2025). JAG1 | NOTCH1 and WNT5A | STAT3 were significant in microglia, representing two independent tumor-to-microglia reprogramming axes – Notch and Wnt-STAT3 signaling are well-characterized mechanisms by which GBM polarizes resident microglia toward an immunosuppressive state (Xu et al., 2025). SFRP2 | FZD6, a Wnt pathway modulator, was significant in CD8-T cells, suggesting tumor-mediated suppression of Wnt signaling in cytotoxic T cells as a potential mechanism of functional impairment. LAMB2 | ITGA1 reinforces the ECM remodeling theme, consistent with laminin-integrin interactions shaping the immunosuppressive niche (Yoshimoto et al., 2025). Together, these interactions define a tumor-centric signaling network that directly reprograms microglia and suppresses CD8-T cell function through ECM, Notch, and Wnt pathway crosstalk.

Discussion

The integrated spatial and single-cell transcriptomic analysis revealed the cellular and molecular architecture of GBM TME at spatial resolution. Three myeloid subpopulations – TAM_APOE, TAM_macrophage, and Monocyte – were identified from scRNA-seq, though Cell2location deconvolution demonstrated that TAM_macrophage maintained a diffuse, low-abundance distribution across the tissue, rarely dominating individual Visium spots. Instead, CD8-T cells emerged as the most spatially structured immune population – forming the tightest self-clusters ($z = 73.95$) while exhibiting the strongest negative co-enrichment with tumor ($z = -51.33$) of any cell pair. Neighborhood enrichment and co-occurrence analyses together confirmed that this pattern reflects active spatial immune exclusion: CD8-T cells are present at high regional abundance but are physically barred from the tumor core at all biologically relevant distances, while tumor cells form a dense self-reinforcing spatial domain. This has direct therapeutic implications – the barrier to cytotoxic immunity in patients is not T cell scarcity but spatially enforced exclusion, suggesting that disrupting the exclusion boundary rather than further T cell recruitment may be required for immunotherapy efficacy (Xu et al., 2025). The ligand-receptor analysis resolved the molecular basis of this architecture: JAG1|NOTCH1 and WNT5A|STAT3 emerged as dominant tumor-to-microglia reprogramming axes, SFRP2|FZD6 highlighted Wnt-mediated suppression of CD8 T cell function, and SPP1|TFRC emerged as the most highly expressed interaction targeting both immune populations — collectively defining a signaling network that simultaneously reprograms resident microglia and impairs cytotoxic T cell function.

It should be noted that preliminary results presented prior to this report were generated from a Cell2location checkpoint trained on an unfinalized annotation and applied to unfiltered spatial spots. Final results reported here were produced by retraining Cell2location on the corrected scRNA-seq annotation and applying deconvolution exclusively to QC-filtered Visium spots (>500 genes, $<25\%$ mitochondrial content), improving both cell-type signature reliability and spatial abundance estimate quality. The spatial localization of these interactions, rather than their mere presence, is the key insight distinguishing this analysis from bulk or single-cell approaches alone. A primary limitation is the use of a single Visium slide from one patient, which constrains spatial generalizability across disease stages. Future work should integrate multiple slides spanning primary and recurrent samples to resolve how the TME evolves with progression. Higher-resolution platforms such as Xenium or MERFISH would further overcome the multi-cell resolution constraint inherent to Visium spot-based capture. Addressing these limitations will be essential for translating spatially-resolved TME findings into clinically actionable therapeutic targets.

Conclusion

This study demonstrates that the GBM tumor microenvironment is defined by spatially structured cellular interactions that cannot be resolved by single-cell sequencing alone. By integrating scRNA-seq with Visium spatial transcriptomics, I identified a spatially enforced immune exclusion phenotype in which CD8 T cells — despite high regional abundance — are physically barred from the tumor core, while tumor cells maintain a dense, self-reinforcing spatial domain. Ligand-receptor analysis identified JAG1|NOTCH1 and WNT5A|STAT3 as tumor-to-microglia reprogramming axes, SFRP2|FZD6 as a Wnt-mediated CD8 T cell suppression signal, and SPP1|TFRC as the dominant immunosuppressive interaction bridging tumor cells to both microglia and CD8 T cells. Together, these spatial and molecular findings converge on a model in which GBM evades cytotoxic immunity not through T cell absence but through spatially organized exclusion reinforced by direct immune reprogramming — a distinction with direct implications for why checkpoint inhibitors have largely failed in GBM despite T cell presence. These findings provide a foundation for spatially-informed therapeutic strategies that target the physical and molecular boundaries enforcing immune exclusion, and highlight the value of multi-modal transcriptomic approaches in dissecting the GBM tumor microenvironment.

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